CERTIFICATE OF COMPLIANCE

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information

CF1R-ALT-02 is applicable to multiple space conditioning systems contained within a single dwelling unit.

01	Project Name:		02	Date Prepared:	
03	Project Location:		04	Building Type:	
05	CA City:		06	Dwelling Unit Name:	
07	Zip Code:		08	Dwelling Unit CFA (ft^2):	
09	Climate Zone:		10	Number of Space Conditioning (SC) Systems in this Dwelling Unit:	

B. Space Conditioning (SC) System Information

01	02	03	04	05	06	07	08	09	10
SC System ID/Name	SC System Description of Area Served	CFA served by this SC System (ft^2):	Is the SC system a ducted system?	Installing a refrigerant containing component?	Installing new SC system components?	Installing more than 25 feet of ducts?	Installing entirely new duct system?	Installing entirely new SC system?	Alteration Type:

C. Extension of Existing Duct System, Greater Than 25 Feet (Section 150.2(b)1Diib)

Required Documentation:

CF2R-MCH-01-E - Space Conditioning Systems

-Duct insulation requirement for the new portions of supply-air and return-air ducts or plenums: R-6 (CZ 3, 5-7and R-8 (CZ1,2,4, 8-16).

CF2R & CF3R-MCH-20-H – Duct Leakage Test

-Leakage rate compliance: less than or equal to 10%, or less than or equal to 7% leakage to outside, or seal all accessible leaks

Exceptions:

Existing duct systems constructed, insulated, or sealed with asbestos are exempt from MCH-20 duct leakage testing requirements

01	02	03	
SC System ID/Name	SC System Description of Area Served	Required New Duct R-Value	

D. Altered Space Conditioning System (Sections 150.2(b)1E and F)

Required Documentation:

CF2R-MCH-01-E - Space Conditioning Systems

-Duct insulation requirement for the new portions of supply-air and return-air ducts or plenums: R-6 (CZ3, 5-7) and R-8 (CZ1,2,4, 8-16)

CF2R & CF3R-MCH-20-H – Duct Leakage Test required when heating or cooling components are installed in ducted systems, or when more than 25 ft of duct length is replaced.

-Leakage rate compliance: less than or equal to 10%, or less than or equal to 7% leakage to outside, or seal all accessible leaks.

CF2R & CF3R-MCH-25-H Refrigerant Charge verification required when refrigerant containing components are installed or altered (applicable in CZ 2, 8-15).

CF2R & CF3R-MCH-23 Airflow Rate greater than or equal to 300 CFM/ton required when MCH-25 is required.

Exceptions:

-Duct systems registered with ~~HERS~~ECC provider as previously sealed are exempt from MCH-20 Duct Leakage Testing requirements.

~~-Heating-only systems and Air Handler/Furnace changes do not require verification of Air Flow MCH-23, or Refrigerant Charge MCH-25.~~

-Existing duct systems constructed, insulated, or sealed with asbestos are exempt from MCH-20 Duct Leakage Testing requirements.

01	02	03	04	05	06	07	08	09	10	10b	11	12	13
SC System ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2/ CEER	Required Thermostat Type	New or Replaced Duct Length	New Duct R-Value

E. Entirely New or Complete Replacement Duct System, with or without Equipment Changeout (Sections 150.2(b)1Diia and 150.2(b)1E, F)

Required Documentation:

CF2R-MCH-01-E - Space Conditioning Systems

-Duct insulation requirement for the new portions of supply-air and return-air ducts or plenums: R-6 (CZ3,5-7) and R-8 (CZ1,2,4, 8-16)

CF2R & CF3R-MCH-20-H Duct Leakage Test required.

-Leakage rate compliance: less than or equal to 5 percent.

CF2R & CF3R-MCH-22 Fan Efficacy

CF2R & CF3R-MCH-23 Airflow Rate

-Compliance: Fan Efficacy less than or equal to 0.58 W/cfm for non-gas furnaces ~~and 0.45 W/cfm for gas furnaces~~ and System Airflow greater than or equal to 350 cfm/ton.

-Alternative Compliance: CF2R & CF3R-MCH-28 Return Duct Design verification is an alternative to MCH-22 and MCH-23 verification.

CF2R & CF3R-MCH-25-H Refrigerant Charge verification required when refrigerant containing components are installed or altered (applicable in CZ 2, 8-15).

~~Exceptions:~~

~~Heating-only systems are exempt from the 0.58 W/cfm and 350 cfm/ton requirements.~~

Note:

An "entirely new or complete replacement duct system" means at least 75 percent of the duct system is new duct material, and up to 25 percent may consist of reused parts from the dwelling unit's existing duct system (e.g., registers, grilles, boots, air handler, coil, plenums, duct material) if the reused parts are accessible and can be sealed to prevent leakage.

01	02	03	04	05	06	07	08	09	10	10b	11	12
SC System Identification or ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2/ CEER	Required Thermostat Type	New Duct R-Value

F. Entirely New or Complete Replacement Space Conditioning System (Section 150.2(b)1C)

Required Documentation:

CF2R-MCH-01-E - Space Conditioning Systems

-Duct insulation requirement for the new portions of supply-air and return-air ducts or plenums: R-6 (CZ3, 5-7) and R-8 (CZ1,2,4, 8-16)

CF2R & CF3R-MCH-20-H Duct Leakage Test required.

-Leakage rate compliance: less than or equal to 5 percent.

CF2R & CF3R-MCH-22 Fan Efficacy

CF2R & CF3R-MCH-23 Airflow Rate

-Compliance: Fan Efficacy less than or equal to 0.58 W/cfm for non-gas furnaces ~~and 0.45 W/cfm for gas furnaces~~ and System Airflow greater than or equal to 350 cfm/ton.

- Alternative Compliance: CF2R & CF3R-MCH-28 Return Duct Design verification is an alternative to MCH-22 and MCH-23 verification.

CF2R & CF3R-MCH-25-H Refrigerant Charge verification required when refrigerant containing components are installed or altered (applicable in CZ 2, 8-15).

~~Exceptions:~~

~~Heating-only systems are exempt from the 0.58 W/cfm and 350 cfm/ton requirements.~~

Note:

An "entirely new or complete replacement duct system" means at least 75 percent of the duct system is new duct material, and up to 25 percent may consist of reused parts from the dwelling unit's existing duct system (e.g., registers, grilles, boots, air handler, coil, plenums, duct material) if the reused parts are accessible and can be sealed to prevent leakage.

01	02	03	04	05	06	07	08	09	10	10b	11	12
SC System ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2/ CEER	Required Thermostat Type	New Duct R-Value

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Compliance documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Signature Date:
Address:	CEA/AEA/ ~~HERS~~ECC Certification Identification (if applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Compliance is true and correct.
2. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer).
3. The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
4. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
5. I understand that a registered copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement.
6. I understand that a registered copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish these requirements.

Responsible Designer Name:	Responsible Designer Signature:
Company:	Date Signed:
Address:	License:
City/State/Zip:	Phone:

CF1R-ALT-02-E User Instructions

Minimum requirements for prescriptive HVAC alteration compliance can be found in Building Energy Efficiency Standards Section 150.2(b)1C.

Completing these forms will require that you have the ~~2022~~ 2025 Reference Appendices for the ~~2022~~ 2025 Building Energy Efficiency Standards.

When the term CF1R is used, it is referencing the CF1R-ALT-02. Worksheets are identified by their entire name, and subsequently by only the worksheet number, such as CF1R-ENV-02.

Instructions for sections with column numbers and row numbers are given separately.

If any part of the alteration does not comply with the prescriptive requirements, prescriptive compliance fails and the performance compliance approach must be used.

A. General Information

1. Project Name: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, enter a unique project identifier such as the house number and street name or example: “Jones’ Furnace Change out.”
2. Date Prepared: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, enter the date of document preparation.
3. Project Location: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, enter the legal street address of property or other applicable identifying information.
4. Building Type: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. ~~If not,~~ this field will automatically default to Single Family.
5. CA City: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, enter the legal city/town of property.
6. Dwelling Unit Name: Enter a unique dwelling unit name or any other identifying name that would readily distinguish this dwelling unit from others in this project.
7. Zip Code: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, enter the 5-digit zip code for the project location (used to determine climate zone).
8. Dwelling Unit CFA (ft^2): If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. For one-dwelling projects, this field will equal the conditioned floor area (CFA) on that document. For multi-dwelling projects, this field will sum with other dwelling units to equal the total CFA on that document. If this project does not utilize a CF1R-ALT-01 (or CF1R-ADD-01), enter the conditioned floor area in ft^2 of the project. If multiple systems are being affected, a CFA value will be assigned to each

system in Section B. Those must sum to this total for the project. For projects NOT involving all systems in the dwelling, this is the CFA of only the portion of the dwelling unit affected.

9. Climate Zone: If the project utilizes a CF1R-ALT-01 (or CF1R-ADD-01), this field will reference the same field on that document for consistency. If not, select the correct climate zone for the project. From the Reference Appendices, Joint Appendix, JA2.1.1.
10. Number of Space Conditioning (SC) Systems in this Dwelling Unit: Enter the number of space conditioning systems in the dwelling unit.

B. Space Conditioning (SC) System Information (Section 150.2(b)1C)

1. SC System Identification or Name: Enter a unique identifier for this system that will readily distinguish it from other systems in the dwelling unit, such as "HVAC1," "upstairs system," etc. It is recommended to mark the system with this identifier using a permanent marker for ease of identification in the field. For single-system dwelling units, enter a simple name such as "HVAC."
2. SC System Description of Area Served: Enter a unique description of the portion of dwelling unit served by this system, such as "entire second floor," "bedroom wing," etc. For single-system dwelling units, enter a simple description such as "entire house."
3. CFA served by this SC System (ft^2): Enter the CFA served by this system.
4. Is the altered or installed system a ducted system? Select "**YES"** if the system has a central air handler (package or split) that is connected to one or more supply air outlets via ducting of any shape or material. Select "**NO**" for nonducted systems such as ductless mini-splits, through-the-wall systems, package terminal air conditioners, etc.
5. Altering or installing a refrigerant containing component? Select "**YES**" if the project includes installing or replacing a component that contains refrigerant; otherwise select "**NO**." Refrigerant containing components include compressors, condensing coils, evaporator coils, refrigerant metering devices or refrigerating lines.
6. Installing new components? Select "**YES"** if new HVAC components such as a packaged unit, condensing unit, cooling/heating coil, or air-handling unit (e.g., furnace), etc. are being installed in the system; otherwise select "**NO**."
7. Installing more than 25 linear feet of new or replacement ducts? Select "**YES**" if the project involves installing more than 25 linear feet of new or replacement ducts; otherwise select "**NO**."
8. Is the entire duct system accessible for sealing and is more than 75 percent of the duct system new or replaced? Select "**YES**" when, upon completion of the project, more than 75 percent of the ducts will be new ducts and/or replaced ducts, AND if at any time during the project all of the ducts are accessible for duct sealing; otherwise select "**NO**." "Accessible" is defined in the Reference Appendices, Joint Appendix, JA1.
9. Are all of the system's components and ducts new (entirely new system) or replaced? Select "**YES**" if the duct system meets the definition of an "Entirely New or Replacement Duct System" and all of the heating and cooling components (furnace, condenser, coil, etc.) are all new or replaced; otherwise select "**NO**."
10. Alteration Type: This field is calculated automatically based on the information entered in previous fields. Alteration types are defined in the Reference Appendices, Joint Appendix, JA1. The alteration type will determine which of the following sections are required by this document.

C. Extension of Existing Duct System, Greater Than 25 Feet (Section 150.2(b)1Diib)

1. System Identification or Name: This field is automatically filled from entries in Section B.
2. SC System Location or Description of Area Served. This field is automatically filled from entries in Section B.
3. Required New Duct R-value: This field is automatically calculated based on the climate zone selected in Section A. It represents the minimum R-value required. The installed R-value shown on the installation certificate (CF2R) must meet or exceed this value.

D. Altered Space Conditioning System (Sections 150.2(b)1E and F)

1. System Identification or Name: This field is automatically filled from entries in Section B.
2. SC System Location or Description of Area Served. This field is automatically filled from entries in Section B.
3. Heating System Type: Select the most appropriate heating system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
4. Altered Heating Component: Select the most appropriate heating system components from the list that are being added or replaced as part of this project. You can select multiple choices if needed. If the type of component being altered does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
5. Heating Efficiency Type: Select the heating efficiency type from the list that is appropriate to the type of system being altered or installed.
6. Heating Minimum Efficiency Value: This field is filled automatically based on selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum efficiency value.
7. Cooling System Type: Select the most appropriate cooling system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
8. Altered Cooling Components: User chooses as many as are applicable: Select the most appropriate cooling system components from the list that are being added or replaced as part of this project. You can select multiple choices if needed. If the type of component being altered does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
9. Cooling Efficiency Type: Select the cooling efficiency type from the list that is appropriate to the type of system being altered or installed.
10. Cooling Minimum Efficiency Value: This field is filled automatically based on CF1R-ALT-02 selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.

10b. Cooling Minimum Efficiency Value: This field is filled automatically based on CF1R-ALT-02 selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.

11. Required Thermostat Type: This field is filled automatically based on selections in previous fields. If “setback” appears here, a setback thermostat meeting the minimum requirements of Section 150.0(i) is required to be installed as part of this project.
12. New or Replaced Duct Length: Select the descriptor that describes the amount of duct, at the completion of the project that is added or replaced as part of this project.
13. New Duct R-value: This field is filled automatically based on the entries in previous fields and the climate zone of the project.

E. Entirely New or Complete Replacement Duct System, with or without Equipment Changeout (Sections 150.2(b)1Diia and 150.2(b)1E, F)

1. System Identification or Name: This field is automatically filled from entries in Section B.
2. SC System Location or Description of Area Served. This field is automatically filled from entries in Section B.
3. Heating System Type: Select the most appropriate heating system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
4. Altered Heating Component: Select the most appropriate heating system components from the list that are being added or replaced as part of this project. You can select multiple choices, if needed. If the type of component being altered does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300
5. Heating Efficiency Type: Select the heating efficiency type from the list that is appropriate to the type of system being altered or installed.
6. Heating Minimum Efficiency Value: This field is filled automatically based on selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.
7. Cooling System Type: Select the most appropriate cooling system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
8. Altered Cooling Components: User chooses as many as that are applicable: Select the most appropriate cooling system components from the list that are being added or replaced as part of this project. You can select multiple choices, if needed. If the type of component being altered does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300
9. Cooling Efficiency Type: Select the cooling efficiency type from the list that is appropriate to the type of system being altered or installed.
10. Cooling Minimum Efficiency Value: This field is filled automatically based on selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.
11. Required Thermostat Type: This field is filled automatically based on selections in previous fields. If “setback” appears here, a setback thermostat meeting the minimum requirements is required to be installed as part of this project.
12. New Duct R-value: This field is filled automatically based on the entries in previous fields and the climate zone of the project.

F. Entirely New or Complete Replacement Space Conditioning System (Section 150.2(b)1C)

1. System Identification or Name: This field is automatically filled from entries in Section B.
2. SC System Location or Description of Area Served. This field is automatically filled from entries in Section B
3. Heating System Type: Select the most appropriate heating system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
4. Altered Heating Component: This field is automatically filled.
5. Heating Efficiency Type: Select the heating efficiency type from the list that is appropriate to the type of system being altered or installed.
6. Heating Minimum Efficiency Value: This field is filled automatically based on selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.
7. Cooling System Type: Select the most appropriate cooling system type from the list. If the type of system to be installed does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
8. Altered Cooling Components (user chooses as many as that are applicable): Select the most appropriate cooling system components from the list that are being added or replaced as part of this project. You can select multiple choices, if needed. If the type of component being altered does not appear on the list, please contact the California Energy Commission Hotline at 800-772-3300.
9. Cooling Efficiency Type: Select the cooling efficiency type from the list that is appropriate to the type of system being altered or installed.
10. Cooling Minimum Efficiency Value: This field is filled automatically based on selections in previous fields. This field represents the minimum efficiency to be installed. The actual installed efficiency may be higher and will be recorded on the Installation Certificate (CF2R). Optional: the user may enter a higher-than-default value for situations where local codes or programs require a higher minimum.
11. Required Thermostat Type: This field is filled automatically based on selections in previous fields. If "setback" appears here, a setback thermostat meeting the minimum requirements is required to be installed as part of this project.
12. New Duct R-value: This field is filled automatically based on the entries in previous fields and the climate zone of the project.

A. General Information

<<This section is required>>

<<Static Text - see the top>>.

01	Project Name:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text>>	02	Date Prepared:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input date format; pick from enumerated list>>
03	Project Location:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text: Street address or alternate applicable description of location >>	04	Building Type:	<< reference from CF1R-ALT-01, or CF1R-ADD-01; ~~if applicable, else~~ allowed ~~default to Single Family;~~ fixed value = Single Family >>
05	CA City:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text: a city name>>	06	Dwelling Unit Name:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text>>
07	Zip Code:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text: pick from enumerated list>>	08	Dwelling Unit CFA (ft^2)	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input integer; xxxxx; >>
09	Climate Zone:	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input text: pick from enumerated list>>	10	Number of space conditioning (SC) systems in this dwelling unit.	<< reference from CF1R-ALT-01, or CF1R-ADD-01 if applicable, else user input: integer, xx>>

B. Space Conditioning (SC) System Information

<< require one row of data to be entered in this table for each of the quantity of space conditioning systems entered in A10>> <<This section is required>>

01	02	03	04	05	06	07	08	09	10
SC System ID/Name	SC System Description of Area Served	CFA served by this SC System (ft^2):	Is the SC system a ducted system?	Installing a refrigerant containing component?	Installing new SC system components?	Installing more than 25 feet of ducts?	Installing entirely new duct system?	Installing entirely new SC system?	Alteration Type:
<<user input: text, max 20 characters do not allow duplicate system names to be used for this dwelling unit>>	<<user input: text, max 20 characters do not allow duplicate descriptions to be used for this dwelling unit (for this MCH-01)>>	<<numeric; xxxx; require the sum of the CFA values entered in this column to be equal to the value for CFA entered in A08>>	user pick from list: "yes"; or "no">>	user pick from list: "yes"; or "no">>	user pick from list: "yes"; or "no">>	user pick from list: "yes"; or "no">>	user pick from list: "yes"; or "no">>	user pick from list: "yes"; or "no">>	<< Calculated field: determine the correct result for "alteration type" for entry in this field by the user responses in B04, B05 , B06, B07, B08, B09 and use of "Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements" (inserted below this section); constrain user input for fields B04-B09 to allow only the available combinations of responses given in the Logic Table in rows a through t; alteration types are: *Extension of Existing Duct System (Section C); *Altered Space Conditioning System (Section D); *Entirely New or Complete Replacement Duct System with or without Equipment Changeout (Section E) *Entirely New or Complete Replacement Space Conditioning System (Section F) * No Alteration Performed >>

Logic Table for Determining Alteration Type and ~~HERS~~ECC Verification Requirements (this table not shown on the completed document)

	1	2	3	4	5	6	7	8	9
	Is the altered or installed system a ducted system?	Altering or installing a refrigerant containing component?	Installing new components? (packaged unit, or condensing unit, or cooling/heating coil, or air-handling unit, etc)	Installing more than 25 linear feet of new or replacement ducts?	Is the entire duct system accessible for sealing, and is more than 75% of the duct system new or replaced?	Are all of the system's components and ducts new or replaced? (entirely new system)	alteration type	~~HERS~~ECC	notes
a	no	yes	no	no	no	no	Altered space conditioning system	RC	e.g. alteration to refrigerant containing component - mini-split or packaged AC
b	no	yes	yes	no	no	no	Altered space conditioning system	RC	e.g. changeout mini-split system component
c	yes	no	yes	no	no	no	Altered space conditioning system	DctLk	e.g. new ducted hydronic fan coil unit or furnace
d	yes	no	yes	yes	no	no	Altered space conditioning system	DctLk	e.g. new furnace + duct alteration
e	yes	yes	no	no	no	no	Altered space conditioning system	RC	e.g. alteration to a refrigerant containing component - split system
f	yes	yes	yes	no	no	no	Altered space conditioning system	RC + DctLk	e.g. changeout refrigerant containing components
g	yes	yes	yes	yes	no	no	Altered space conditioning system	RC + DctLk	e.g. changeout refrigerant containing component + altered ducts
h	yes	yes	no	yes	no	no	Altered space conditioning system	RC + DctLk	e.g. alteration to refrigerant containing component + altered ducts
i	yes	no	no	yes	yes	no	Entirely new duct system with or without Equipment Changeout	DctLk + FE/AF or Tbl150.0-C,D	e.g. new duct system without equipment changeout
j	yes	no	yes	yes	yes	no	Entirely new duct system with or without Equipment Changeout	DctLk + FE/AF or Tbl150.0-C,D	e.g. new furnace + new duct system
k	yes	yes	no	yes	yes	no	Entirely new duct system with or without Equipment Changeout	RC + DctLk + FE/AF or Tbl150.0-C,D	e.g. alteration to a refrigerant containing component + new duct system
l	yes	yes	yes	yes	yes	no	Entirely new duct system with or without Equipment Changeout	RC + DctLk + FE/AF or Tbl150.0-C,D	e.g. changeout refrigerant containing component + new duct system

m	no	no	yes	no	no	yes	Entirely new space conditioning system	none	e.g. entirely new ductless hydronic heating system (boiler heating only); or new wall heater
n	no	yes	yes	no	no	yes	Entirely new space conditioning system	RC	e.g. new mini-split (weigh-in); or new room packaged AC (factory charged)
o	yes	no	yes	yes	yes	yes	Entirely new space conditioning system	DctLk	e.g. new ducted hydronic heating system, or other new ducted heating-only system
p	yes	yes	yes	yes	yes	yes	Entirely new space conditioning system	RC + DctLk + FE/AF or Tbl150.0-C,D	e.g. new split system
q	yes	no	no	yes	no	no	Extension of an existing duct system	DctLk	e.g. altered ducts
r	no	no	no	no	no	no	System is exempt from the alteration requirements	none	no alteration performed
s	yes	no	no	no	no	no	System is exempt from the alteration requirements	none	no alteration performed
t	yes	yes	yes	no	yes	yes	Entirely new space conditioning system	RC + DctLk + FE/AF or Tbl150.0-C,D	e.g. new ducted system that has less than 25 ft of ducts
u	no	no	yes	no	no	no	Altered space conditioning system	none	e.g. new ducted hydronic fan coil unit or new hydronic heating boiler

Nomenclature:
RC = Refrigerant Charge Verification (MCH-25) applicable in CZ 2, 8-15
DctLk = Duct Leakage Test (MCH-20)
FE/AF or Tbl150.0-C,D - Fan Efficacy and Airflow Rate verification (MCH-22; MCH-23) or alternative compliance: (MCH-28)

C. Extension of Existing Duct System, Greater Than 25 Feet (Section 150.2(b)1Diib)

<<if there are no Alteration Types in column B10 equal to "Extension of Existing Duct System (Section C)" then display the "section does not apply" message, else require one row of data to be entered in this table for each SC System of alteration type in column B10 equal to: "Extension of Existing Duct System (Section C)"

01	02	03
SC System ID/Name	SC System Description of Area Served	Required New Duct R-Value
<< auto filled from B01 >>	<< auto filled from B02 >>	<<calculated field: if A09= CZ3, 5-7, then value = "R-6."; else if A09=CZ 1, 2, 4, 8-16 then value = "R-8."

<<Static Text - see the top>>

D. Altered Space Conditioning System (Sections 150.2(b)1E and F)

<<**if** there are no Alteration Types in column B10 equal to "Altered Space Conditioning System (Section D)" then display the "section does not apply" message; **else** require one row of data in this table for each SC System of alteration type in column B10 equal to: "Altered Space Conditioning System (Section D)">>

01	02	03	04	05	06	07	08	09	10	10b	11	12	13
SC System ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2/ CEER	Required Thermostat Type	New or Replaced Duct Length	New Duct R-Value
<<auto filled from B01 >>	<< auto filled from B02 >>	<<user pick one from list: *central gas furnace; *central split HP; *central packaged HP; *central large packaged HP; *ductless mini-split HP; *room HP; *boiler; *hydronic; *combined hydronic; *hydronic+forced air; *combined hydronic+forced air; *hydronic HP, *hydronic HP+forced air; *gas wall furnace; *gas space heater; *electric; *non-air-source heat pump; *Wood Heat; * no heating; *Small duct high velocity HP; *Ductless VRF HP; *Packaged gas furnace; *multisplit HP-ducted;	<< user pick as many as are applicable from list: *gas furnace AHU; *fancoil AHU; *outdoor condensing unit; *indoor coil; *boiler; *TXV or EXV; *compressor; *refrigerant lineset; *no heating component altered>>	<< if D04= no heating component altered, then value =n/a, else user pick from list: *AFUE; *HSPF; *HSPF2; *COP>>	<< if D04= no heating component altered, then value =n/a, else user enter numeric value xx.x; default minimum value for AFUE= 80; or default minimum value for HSPF= 8.0; or default minimum value for HSPF2=6. 7allow user to overwrite default value, but flag non-default values and report in	<<user pick one from list: *central split AC; *central split HP; *central packaged AC ; *central packaged HP; * central large packaged AC ; *central large packaged HP; *ductless mini-split AC; *ductless mini-split HP; *gas absorption AC; *room AC; *room HP; *hydronic HP; *hydronic HP+forced air *evaporative – direct; *evaporative – indirect; *evaporative – indirect/direct; *evaporatively cooled condenser; *Ice Storage AC; *non-air-source heat pump; *non-air-cooled air conditioner; *no cooling	<<user pick as many as are applicable from list: *outdoor condensing unit; *indoor fancoil AHU; *indoor coil; *TXV or EXV; *Compressor; *refrigerant lineset; *no cooling component altered>>	<< if D08= no cooling component altered, then value =n/a, else user input. Allowed value = (1) EER/SEER, (2) EER2/SEER2, (3) CEER, (4) EER, or (5) EER2. >>	<< if D08= no cooling component altered, then value =n/a, else user enter value: xx.x; default minimum value for EER/SEER=14;or default minimum value for EER2/SEER2=13.4 allow user to overwrite default value, but flag non-default values and report in project status	<<if D08 = no cooling component altered, then value = N/A; else user enter value: xx.x; default minimum value for EER and EER/SEER=11.7; or default minimum value for EER2 and EER2/SEER2 =11.2; or default minimum value for CEER=8.7; allow user to overwrite default value, but flag non-default values and report in project status notes field>>	<< setback>>	<< calculated field: if B04=no, then display N/A; else user pick from list: *≤25ft; *>25ft; *N/A-no ducts replaced>>	<<calculated field: if B04=no, then display N/A; elseif D12=N/A, then display N/A; elsif A09= CZ3, 5-7, then "R-6."; else if A09=CZ 1, 2, 4, 8-16then "R-8."

		*multisplit HP-ductless; *multisplit HP-ducted+ductless; *VCHP-ducted; *VCHP-ductless; *VCHP-ducted+ductless; *ducted mini-split HP; *SPVHP; *PTHP>>			project status notes field >>	*Small duct high velocity HP; *Small duct high velocity AC; *Ductless VRF HP; *Ductless VRF AC; *multisplit AC-ducted; *multisplit AC-ductless; *multisplit AC-ducted+ductless; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *VCHP-ducted; *VCHP-ductless; *VCHP-ducted+ductless; *ducted mini-split AC; *ducted mini-split HP; *SPVAC; *SPVHP; *PTAC; *PTHP>>			notes field >>				

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E. Entirely New or Complete Replacement Duct System, with or without Equipment Changeout (Sections 150.2(b)1Diia and 150.2(b)1E, F)

<<if there are no Alteration Types in column B10 equal to "Entirely New or Complete Replacement Duct System with or without Equipment Changeout (Section E)" then display the "section does not apply" message; else require one row of data in this table for each SC System of alteration type in column B10 equal to: "Entirely New or Complete Replacement Duct System with or without Equipment Changeout (Section E)">>

01	02	03	04	05	06	07	08	09	10	10b	11	12
SC System Identification or ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER~~/~~SEER2	Cooling Minimum Efficiency Value EER/EER2/CEER	Required Thermostat Type	New Duct R-Value
<< auto filled from B01 >>	<< auto filled from B02 >>	<<user pick one from list: *central gas furnace; *central split HP; *central packaged HP; *central large packaged HP *ductless split HP; *room HP; *boiler; *hydronic; *combined hydronic; *hydronic+forced air; *combined hydronic+forced air; *hydronic HP, *hydronic HP+forced air; *gas wall furnace; *gas space heater; *electric * no heating; *Wood Heat; *Packaged gas furnace; *non-air-source HP; *Small duct high velocity HP;	<< user pick as many as are applicable from list: *gas furnace AHU; *fancoil AHU; *outdoor condensing unit; *indoor coil; *boiler; *TXV or EXV; *compressor; *refrigerant lineset; *no heating component altered>>	<<if E04= no heating component altered, then value =n/a, else user pick from list: *AFUE; *HSPF; *HSPF2; *COP>>	<< if E04= no heating component altered, then value =n/a, elseuser enter value: xx.x; default minimum value for AFUE= 80; or default minimum value for HSPF= 8.0; or default minimum value for HSPF2=6.7; allow user to overwrite default value, but flag non-default values and report in project status notes field >>	<<user pick one from list: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC ; *central large packaged HP *ductless split AC; *ductless split HP; *gas absorption AC *room AC; *room HP; *hydronic HP, *hydronic HP+forced air; *evaporative - direct; *evaporative – indirect; *evaporative - indirectdirect *evaporatively cooled condenser; *Ice Storage AC; *no cooling; *Ductless VRF HP; *Ductless VRF AC;	<<user pick as many as are applicable from list: *outdoor condensing unit, *indoor fancoil AHU, *indoor coil, *TXV or EXV, *Compressor, *refrigerant lineset, *no cooling component altered>>	<< if E08= no cooling component altered, then value =n/a, else user input. Allowed value = (1) EER/SEER, (2) EER2/SEER2, (3) CEER, (4) EER, or (5) EER2.>>	<< if E08= no cooling component altered, then value =n/a else user enter value: xx.x; default minimum value for EER/SEER=14 ; or default minimum value for EER2/SEER2= 13.4; allow user to overwrite default value, but flag non-default values and report in project status notes field >>	<<if E08 = no cooling component altered, then value = N/A; else user enter value: xx.x; default minimum value for EER and EER/SEER=11.7; or default minimum value for EER2 and EER2/SEER2=11.2; or default minimum value for CEER=8.7; allow user to overwrite default value, but flag non-default values and report in project status notes field>>	<<setback>>	<<calculated field: if A09= CZ3, 5-7, then "R-6."; else if A09=CZ 1, 2, 4, 8-16then "R-8."

		*Ductless VRF HP; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *ducted mini-split HP; *SPVHP; *PTHP >>				*multisplit AC-ducted; *multisplit AC-ductless; *multisplit AC-ducted+ductless; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *non-air-source heat pump; *non-air-cooled air conditioner; *Small duct high velocity HP; *Small duct high velocity AC; *ducted mini-split AC; *ducted mini-split HP; *SPVAC; *SPVHP; *PTAC; *PTHP>>						

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F. Entirely New or Complete Replacement Space Conditioning System (Section 150.2(b)1C)

<<if there are no Alteration Types in column B10 equal to "Entirely New or Complete Replacement Space Conditioning System (Section F)" then display the "section does not apply" message; else require one row of data in this table for each SC System of alteration type in column B10 equal to: "Entirely New or Complete Replacement Space Conditioning System (Section F)">>

01	02	03	04	05	06	07	08	09	10	10b	11	12
SC System ID/Name	SC System Description of Area Served	Heating System Type	Altered Heating Component	Heating Efficiency Type	Heating Minimum Efficiency Value	Cooling System Type	Altered Cooling Components	Cooling Efficiency Type	Cooling Minimum Efficiency Value SEER/SEER2	Cooling Minimum Efficiency Value EER/EER2 /CEER	Required Thermostat Type	New Duct R-Value
<< auto filled from B01 >>	<< auto filled from B02 >>	<<user pick one from list: *central gas furnace; *central split HP; *central packaged HP *central large packaged HP *ductless split HP; *room HP; *boiler; *hydronic; *hydronic+forced air; *combined hydronic+forced air; *hydronic HP, *hydronic HP+forced air; *combined hydronic; *gas wall furnace; *gas space heater; *electric * no heating; *Wood Heat; *Packaged gas furnace; *non-air-source HP; *Small duct high velocity HP; *Ductless VRF HP; *multisplit HP-ducted; *multisplit HP-ductless;	<< calculated field: **if** F03= no heating; **then** textvalue= no heating component altered; **else** value=entirely new heating system>>	<< if F04= no heating component altered, then value =n/a; else user pick from list: *AFUE; *HSPF; *HSPF2; *COP>>	<<if F04= no heating component altered, then value =n/a; else user enter value: xx.x; default minimum value for AFUE= 80; or default minimum value for HSPF= 8.0; or default minimum values for HSPF2=6.7; allow user to overwrite default value, but flag non-default values and report in project status notes field >>	<<user pick one from list: *central split AC; *central split HP *central packaged AC ; *central packaged HP *central large packaged AC; *central large packaged HP *ductless split AC; *ductless split HP; *gas absorption AC *room AC; *room HP; *hydronic HP, *hydronic HP+forced air; *evaporative – direct; *evaporative – indirect; *evaporative – indirectdirect; *evaporatively cooled condenser; *Ice Storage AC; *no cooling;	<< calculated field: if F07=no cooling; then value= no cooling component altered; else value= entirely new cooling system>>	<< **if** F08= no cooling component altered, **then** value =n/a, **else** user input. Allowed value = (1) EER/SEER, (2) EER2/SEER2, (3) CEER, (4) EER, or (5) EER2.>>	<< if F08= no cooling component altered, then value =n/a, else user enter numeric value: xx.x; default minimum value for EER/SEER=14; or default minimum value for EER2/SEER2=13.4; allow user to overwrite default value, but flag non-default values and report in project status notes field >>	<<if F08=no cooling component altered, then value = N/A; else user enter numeric value: xx.x; default minimum value for EER and EER/SEER= 11.7; or default minimum for EER2 and EER2/SEER2=11.2; or default minimum value for CEER=8.7; allow user to overwrite default value, but flag non-default values and report in project status notes field>>	< setback>>	<<calculated field: **if** B04=no, then display text value= N/A; **elsif** A09= CZ 3, 5-7 **then value=** "R-6."; **else** if A09=CZ 1, 2, 4, 8-16 **then** value= "R-8."

		*multisplit HP-ducted+ductless; *ducted mini-split HP; *SPVHP; *PTHP >>				*Ductless VRF HP; *Ductless VRF AC; *multisplit AC-ducted; *multisplit AC-ductless; *multisplit AC-ducted+ductless; *multisplit HP-ducted; *multisplit HP-ductless; *multisplit HP-ducted+ductless; *non-air-source heat pump; *non-air-cooled air conditioner; *Small duct high velocity HP; *Small duct high velocity AC; *ducted mini-split AC; *ducted mini-split HP; *SPVAC; *SPVHP; *PTAC; *PTHP >>						
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